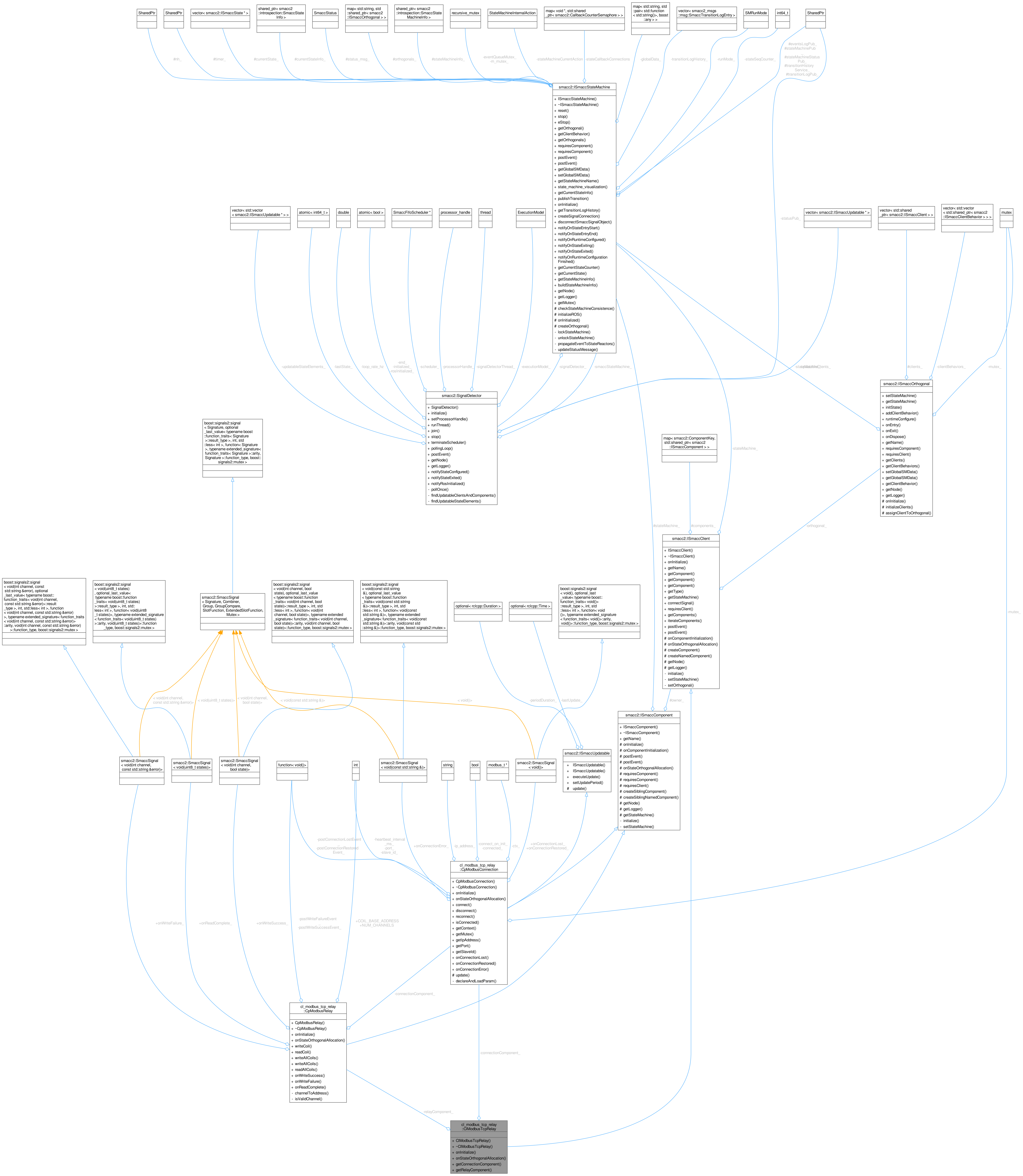
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
#nh_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#eventsLogPub_
#stateMachinePub
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
-statusPub_
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
mutex
-updatableStateElements_
-lastState_
-loop_rate_hz
-end_
-initialized_
-rosInitialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-signalDetector_
-smaccStateMachine_
-stateMachine_
-updatableClients_
#clients_
-clientBehaviors_
-mutex_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
#stateMachine_
#components_
-orthogonal_
-mutex_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < void(int channel, const std::string &error), optional _last_value< typename boost:: function_traits< void(int channel, const std::string &error)>::result _type >, int, std::less< int >, function < void(int channel, const std::string &error) >, typename extended_signature< function_traits < void(int channel, const std::string &error)> ::arity, void(int channel, const std::string &error) >::function_type, boost::signals2::mutex >
boost::signals2::signal < void(uint8_t states) , optional_last_value< typename boost::function _traits< void(uint8_t states) >::result_type >, int, std:: less< int >, function< void(uint8 _t states)>, typename extended_signature < function_traits< void(uint8_t states) >::arity, void(uint8_t states)>::function _type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(int channel, bool state), optional_last_value < typename boost::function _traits< void(int channel, bool state)>::result_type >, int, std ::less< int >, function< void(int channel, bool state)>, typename extended _signature< function_traits< void(int channel, bool state)>::arity, void(int channel, bool state)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const std::string &), optional_last_value < typename boost::function _traits< void(const std::string &)>::result_type >, int, std ::less< int >, function< void(const std::string &)>, typename extended _signature< function_traits< void(const std::string &)>::arity, void(const std ::string &)>::function_type, boost::signals2::mutex >
optional< rclcpp::Duration >
optional< rclcpp::Time >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
< void(int channel, const std::string &error)>
< void(uint8_t states)>
< void(int channel, bool state)>
< void(const std::string &)>
< void()>
-periodDuration_
-lastUpdate_
#owner_
smacc2::SmaccSignal < void(int channel, const std::string &error)>
smacc2::SmaccSignal < void(uint8_t states)>
smacc2::SmaccSignal < void(int channel, bool state)>
function< void()>
int
smacc2::SmaccSignal < void(const std::string &)>
string
bool
modbus_t *
smacc2::SmaccSignal < void()>
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
-postConnectionLostEvent
-postConnectionRestored Event_
-heartbeat_interval _ms_
-port_
-slave_id_
+onConnectionError_
-ip_address_
-connect_on_init_
-connected_
-ctx_
+onConnectionLost_
+onConnectionRestored_
cl_modbus_tcp_relay ::CpModbusConnection
+ CpModbusConnection()
+ ~CpModbusConnection()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ connect()
+ disconnect()
+ reconnect()
+ isConnected()
+ getContext()
+ getMutex()
+ getIpAddress()
+ getPort()
+ getSlaveId()
+ onConnectionLost()
+ onConnectionRestored()
+ onConnectionError()
# update()
- declareAndLoadParam()
+onWriteFailure_
+onReadComplete_
+onWriteSuccess_
-postWriteFailureEvent
-postWriteSuccessEvent_
+COIL_BASE_ADDRESS
+NUM_CHANNELS
-connectionComponent_
cl_modbus_tcp_relay ::CpModbusRelay
+ CpModbusRelay()
+ ~CpModbusRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ writeCoil()
+ readCoil()
+ writeAllCoils()
+ writeAllCoils()
+ readAllCoils()
+ onWriteSuccess()
+ onWriteFailure()
+ onReadComplete()
- channelToAddress()
- isValidChannel()
-connectionComponent_
-relayComponent_
cl_modbus_tcp_relay ::ClModbusTcpRelay
+ ClModbusTcpRelay()
+ ~ClModbusTcpRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ getConnectionComponent()
+ getRelayComponent()